Complete benchmark and finer annotation tasks

a

	PBMC coarse			PBMC fine			Cortex development			Cortex held studies		
	1	10	100	1	10	100	1	10	100	1	10	100
CellPFN	60.1	84.0	91.9	**51.1**	79.0	89.0	41.4	**63.6**	**66.1**	62.8	**91.9**	**94.7**
TabICLv2	50.2	81.4	92.4	30.0	73.8	83.6	39.5	59.9	65.3	49.8	83.2	92.9
TabICLv2 + prior	62.3	**85.2**	**93.4**	29.1	64.6	85.3	44.6	60.6	65.3	62.7	85.8	93.2
Kumo small	45.8	82.2	92.6	44.1	80.0	88.3	31.2	57.5	65.1	48.8	84.8	93.7
Kumo small + prior	**62.8**	83.3	92.3	49.0	80.5	88.7	41.9	59.2	64.9	59.3	86.8	92.9
Kumo large	58.9	81.7	92.8	46.3	**83.3**	**89.3**	40.1	56.8	65.1	49.1	85.1	93.1
Kumo large + prior	62.2	82.5	92.8	50.9	82.2	89.2	42.8	60.6	65.8	60.8	85.9	93.4
TabPFN-3.5	34.3	82.6	91.5	33.0	79.3	86.9	38.4	59.3	65.5	54.1	85.3	93.9
Geneformer + LR	**67.1***	**88.5***	92.7*	**68.8***	**84.9***	**90.4***	43.1(*)	61.1(*)	65.4(*)	58.4*	87.4*	90.3*
scGPT + LR	64.9*	77.5*	80.7*	55.7*	65.9*	72.0*	**47.6**	60.2	65.8	**65.5**	87.2	89.5
Low-rank LDA	—	79.4	90.0	—	70.3	87.8	—	60.6	66.0	—	91.0	**94.7**
CellTypist	39.8	77.5	87.8	33.9	69.7	84.2	34.9	57.8	63.9	44.4	87.8	91.4
SingleR	56.7	78.2	85.9	46.0	68.6	83.3	44.4	61.7	63.9	56.8	86.6	89.1
scANVI inductive	20.0	40.7	86.7	21.7	51.8	83.9	—	—	—	—	—	—
scANVI transductive	40.5	69.1	89.1	40.8	70.8	86.2	—	—	—	—	—	—
Logistic regression	37.5	80.5	91.7	39.0	73.4	85.1	34.7	57.4	65.5	45.0	73.5	90.6

Supports per class

b Finer tasks: CellPFN, up to 8,192 HVGs

	1	10	100
CD4 T, level 2 (4)	36.3	69.2	85.6
T, level 2 (9)	27.4	64.4	83.8
CD4 T, level 3 (9)	23.4	56.3	79.6
CD8 T, level 3 (7)	39.5	69.4	84.4
B, level 3 (6)	31.6	63.6	79.8
NK, level 3 (6)	36.2	64.3	73.0
T, level 3 (17)	25.0	55.7	76.8
All PBMC, level 3 (35)	39.8	68.4	78.4
Cortex finer types (7)	67.8	76.3	79.3
Held-study neural (3)	52.5	62.1	69.6

Supports per class

c Kidney, 100 supports/class

	5/A	5/B	9/A	9/B
CellPFN, counts	96.0	96.2	97.2	97.3
CellPFN, processed	94.7	94.8	96.0	96.1
TabICLv2	**97.7**	**97.2**	**97.9**	**97.9**
Kumo large	97.5	96.8	97.7	97.5
Geneformer + LR	**97.7***	**97.8***	**98.2***	**98.0***
scGPT + LR	94.0	94.2	95.1	95.1
Low-rank LDA	96.0	96.0	96.9	96.9

Classes / feature panel

d Exposure and comparison boundaries

* PBMC: likely query-cell exposure for Geneformer and scGPT.
* Held cortex: Geneformer source studies cover 92.6% of query cells.
(*) Development cortex: possible Geneformer overlap, at most 22.3%. Exact-cell and target-label matches unverified.

Values: balanced accuracy (%). Gray * / (*) values: query-source exposure; bold: best overall and best unflagged, including ties.
Finer tasks are post hoc and do not constitute independent cohorts. TabPFN-3.5: non-commercial research use only.